Analyzing Personal Performance and Patterns in the Rhythm Game osu: Insights from Gameplay Data

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***Abstract*—Abstract— This study investigates factors influencing player performance in *osu!* by analyzing 301 gameplay sessions from a single participant over one week. Key gameplay metrics (accuracy, max combo, misses) and beatmap features (difficulty, object density, type) were collected and processed, with derived metrics such as combo ratio and miss rate. Statistical analyses, including Pearson correlation, Mann–Whitney U tests, and one-way ANOVA, were conducted to explore relationships between these variables and combo ratio. Results indicate that higher accuracy, sustained combos, and prior map familiarity positively affect performance, while errors, denser maps, and higher difficulty reduce combo ratio. Beatmap type and music genre also significantly influence outcomes, whereas temporal factors have minimal effect. These findings highlight how gameplay and map characteristics shape performance, providing actionable insights for players seeking to optimize rhythm game outcomes.**

***Index Terms*—*osu!*, gameplay analysis, rhythm games, data science, performance metrics, combo ratio, object density, miss rate, statistical analysis, visualization.**

I. Introduction

Personal data tracking, often referred to as the "quantified self," enables individuals to gain insights into habits, skills, and performance through systematic logging. Extending this concept to gaming behavior allows players to analyze their own performance, identify patterns, and understand factors influencing success in complex interactive tasks.

Rhythm games, such as *osu!*, require precise timing, motor coordination, and sustained attention, and they record detailed gameplay metrics, including accuracy, combo counts, and object interactions. Prior research indicates that rhythm game practice can improve attention, processing speed, short-term memory, and sensorimotor coordination [1]–[3], suggesting that gameplay metrics such as accuracy and combo counts are meaningful indicators of skill. Despite the popularity of rhythm games, few longitudinal studies examine individual gameplay performance using detailed play-level data. This study addresses this gap by analyzing 301 personal sessions in *osu!*, providing an opportunity to explore how gameplay metrics and beatmap characteristics influence combo ratio over time. Here, combo ratio is defined as the ratio of achieved combo to maximum possible combo on a beatmap.

A. Research Objectives

The objectives of this study are as follows:

1) To identify gameplay metrics that significantly influence player performance in *osu!*, as measured by combo ratio.
2) To examine the relationships between beatmap characteristics and combo ratio.
3) To determine differences in combo ratio across various beatmap types.
4) To investigate temporal patterns in combo ratio and their effects on sustained player performance.
5) To highlight the key factors that most strongly affect sustained combo performance.

B. Research Questions

The objectives of this study are as follows:

- What correlations exist between combo ratio and gameplay metrics (Accuracy, Miss, Passed, etc)?
- How does beatmap features (difficulty rating, approach rate, circle size, etc) correlate with combo ratio?
- How does combo ratio differ across beatmap types (Hybrid, Jump, Stream, Tech)?
- How does combo ratio vary over time, such as by hour of day, day of the week, or sequence of sessions?

II. Literature Review

Rhythm games, such as *osu!*, require precise timing, motor coordination, and sustained attention, making them an ideal context for studying human cognitive-motor performance. Prior studies have demonstrated that regular rhythm game practice can enhance selective attention, processing speed, short-term memory, and sensorimotor coordination [1]–[3].

These findings suggest that rhythm games provide measurable cognitive and motor benefits, making gameplay metrics such as accuracy and combo counts relevant for experimental investigation.

While large-scale game analytics examines session-level gameplay metrics to identify patterns in player behavior [4], EEG-based studies have shown that neural signals can provide additional insights into engagement and cognitive performance during gameplay [5].

Experimental studies, such as Kihara and Hagiwara [1], combined rhythm game sessions with cognitive testing and EEG recordings to assess attentional functions, confirming correlations between gameplay metrics and cognitive outcomes. Other EEG-based and neuroimaging studies similarly indicate

TABLE I: Dataset Variables, Beatmap Characteristics, and Gameplay Metrics

Variable	Type	Scale / Unit
Metadata and Identifiers		
id	Quantitative	Integer
song_title	Categorical	–
artist	Categorical	–
genre	Categorical	–
beatmap_id	Quantitative	Integer
date_started	Timestamp	YYYY-MM-DD HH:MM:SS
date_ended	Timestamp	YYYY-MM-DD HH:MM:SS
Beatmap Characteristics		
beatmap_type	Categorical	–
difficulty_rating	Quantitative	Stars
bpm	Quantitative	Beats per minute
hit_length	Quantitative	Seconds
count_circles	Quantitative	Count
count_sliders	Quantitative	Count
count_spinners	Quantitative	Count
ar	Quantitative	Approach rate
cs	Quantitative	Circle size
Gameplay Performance Metrics		
accuracy	Quantitative	%
max_combo_player	Quantitative	Count
max_combo_beatmap	Quantitative	Count
max_great	Quantitative	Count
great	Quantitative	Count
ok	Quantitative	Count
meh	Quantitative	Count
miss	Quantitative	Count
is_perfect_combo	Categorical	True/False
passed	Categorical	True/False
played_before	Categorical	True/False

TABLE II: Dataset Variables, Beatmap Characteristics, and Gameplay Metrics with Descriptions

Variable	Description
Metadata and Identifiers	
id	Unique session identifier (Integer)
song_title	Name of the song played
artist	Song artist
genre	Genre of the song
beatmap_id	Unique beatmap identifier
date_started	Timestamp of session start
date_ended	Timestamp of session end
Beatmap Characteristics	
beatmap_type	Beatmap category (Hybrid, Jump, Stream, Tech)
difficulty_rating	Beatmap difficulty rating in stars
bpm	Speed of the beatmap in beats per minute
hit_length	Duration of the beatmap in seconds
count_circles	Number of hit circles in the beatmap
count_sliders	Number of sliders in the beatmap
count_spinners	Number of spinners in the beatmap
ar	Approach rate of the beatmap
cs	Circle size of the beatmap
Gameplay Performance Metrics	
accuracy	Player's accuracy for the session in %
max_combo_player	Maximum combo achieved by the player
max_combo_beatmap	Maximum possible combo of the beatmap
is_perfect_combo	Whether a perfect combo was achieved
passed	Whether the player passed the beatmap
max_great	Maximum great hits for the session
great	Number of great hits
ok	Number of ok hits
meh	Number of meh hits
miss	Number of misses
played_before	Whether the beatmap was previously played by the player

that engagement and performance in cognitive games are measurable through neural signals [6].

While these studies provide valuable insights, they primarily rely on aggregated population-level data or short-term laboratory experiments, limiting understanding of longitudinal, individual-level performance.

To address this gap, the present study analyzes 301 sessions from a single player, integrating raw gameplay metrics and derived features such as combo ratio, object density, and miss rate. Unlike prior work that emphasizes population-level trends or predictive modeling, this study focuses on personal gameplay data to examine how beatmap characteristics and gameplay features influence individual performance over time. By providing detailed, longitudinal insights, this work complements existing rhythm game research while highlighting individual player performance patterns that are typically overlooked in larger-scale analyses.

III. Methodology

A. Participants

The research was carried out with a single participant, who was also the researcher, serving as the sole subject of the study. This participant is a 22-year-old fourth-year student at National University – Manila, pursuing a Bachelor of Science in Computer Science with a specialization in Machine Learning. The participant is an experienced *osu!* player, with prior familiarity in rhythm games, and consistently engages in gameplay sessions spanning various beatmap types to maintain and develop skill.

Throughout the study, gameplay data were self-recorded by the participant over multiple sessions, covering a comprehensive range of performance metrics. Data collection was conducted entirely non-invasively, utilizing the *osu!* API in combination with manual logging of play-specific details. No personal or sensitive information beyond the participant's gameplay performance was captured during the process, ensuring that the dataset remained focused strictly on observable gameplay outcomes.

B. Data Collection Methods

Gameplay data were recorded for each play using the *osu!* API, supplemented with manual logging of play-specific details. The dataset was gathered over a one-week period and comprises 301 valid gameplay sessions. Tracked variables include beatmap metadata, beatmap characteristics, and gameplay performance metrics. All sessions were documented immediately after each play attempt to ensure accuracy and consistency. Only non-sensitive performance metrics were collected, and all recorded data were stored in a CSV File for subsequent cleaning, analysis, and visualization. See Table I.

C. Operational Definitions

Table II summarizes the variables analyzed in this study.

Beatmap refers to a level or map in *osu!* that defines the placement, timing, and type of notes the player must hit, essentially representing the song's playable structure. **Beatmap**

Type is a categorical variable describing the structure and design of an *osu!* map, with the following categories:

- **Jump:** Spaced-out notes requiring large cursor movements.
- **Stream:** Dense sequences of notes requiring rapid, continuous clicking.
- **Tech:** Complex rhythms and patterns requiring precise slider and circle execution.
- **Hybrid:** Combines multiple map styles and note patterns.

D. Data Preprocessing

Data Preprocessing was performed to ensure consistency, accuracy, and usability of the dataset for analysis. The following steps were applied:

- **Derived Features:**
  - `combo_ratio` – Calculated as `max_combo_player/max_combo_beatmap`, representing the player's proportion of the maximum possible combo.
  - `total_objects` – Calculated as `count_circles + count_sliders + count_spinners`, representing the total interactive elements in a beatmap.
  - `object_density` – Calculated as `total_objects/hit_length`, indicating the number of objects per second in the beatmap.
  - `miss_rate` – Calculated as `miss/total_objects`, providing a normalized measure of missed notes relative to total objects.
- **Missing Values:** The dataset originally contained missing entries in `meh` (57 missing) and `miss` (3 missing). These were imputed with zeros, as missing hits typically indicate no occurrence rather than a recording error. All other columns were complete.
- **Categorical Standardization:** The `beatmap_type` column was reviewed to ensure consistency. Minor inconsistencies in spelling, capitalization, and spacing were corrected, resulting in four uniform categories: Hybrid, Jump, Stream, and Tech.
- **Temporal Data:** `day` and `hour` were extracted from session timestamps (`date_started`) to allow chronological sorting and temporal analysis.
- **Outliers:** No major outliers were removed, as all values were within plausible ranges. Extreme values (e.g., very high BPM or combo counts) were preserved because they reflect actual gameplay performance.

After cleaning, the final dataset contained 301 complete sessions with 27 columns, including both raw and derived features, ready for exploratory and statistical analysis.

E. Statistical Analysis

Statistical analysis was conducted to explore relationships and patterns in gameplay performance, focusing on `combo_ratio`. The following techniques were applied:

- **Pearson Correlation:** To assess linear relationships between `combo_ratio` and numerical features.

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}} \tag{1}$$

  where x_i and y_i are individual observations, and $\bar{x}$ and $\bar{y}$ are the respective means.
  Hypotheses:
  - Null hypothesis (H_0): There is no linear correlation between the feature and `combo_ratio` ($r = 0$).
  - Alternative hypothesis (H_1): There is a linear correlation between the feature and `combo_ratio` ($r \neq 0$).
- **Mann–Whitney U Test:** To compare the distribution of `combo_ratio` between two groups, e.g., previously played beatmaps and newly played beatmaps. The Mann-Whitney U statistic is computed as:

$$U = \min(U_1, U_2), \tag{2}$$

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1, \tag{3}$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2 \tag{4}$$

  where n_1, n_2 are the sample sizes of the two groups, and R_1, R_2 are the sums of ranks for each group.
  Hypotheses:
  - Null hypothesis (H_0): The distributions of `combo_ratio` are the same for both groups.
  - Alternative hypothesis (H_1): The distributions of `combo_ratio` differ between the groups.
- **One-Way ANOVA:** To test for significant differences in `combo_ratio` across multiple beatmap types (Hybrid, Jump, Stream, Tech). Assumes normality and homogeneity of variance.

$$F = \frac{\sum_i n_i(\bar{x}_i - \bar{x})^2/(k-1)}{\sum_i \sum_j (x_{ij} - \bar{x}_i)^2/(N-k)} \tag{5}$$

  where k is the number of groups, N is the total number of observations, n_i is the group size, and $\bar{x}_i$ is the group mean.
  Hypotheses:
  - Null hypothesis (H_0): The mean `combo_ratio` is the same across all groups.
  - Alternative hypothesis (H_1): At least one group's mean `combo_ratio` differs from the others.
- All analyses were performed in Python using `pandas` and `scipy.stats` in VScode. Statistical significance was evaluated at $p < 0.05$, and visualizations such as histograms, boxplots, scatterplots, and heatmaps supported interpretation of trends and relationships.

IV. Results

This section presents the results of the analysis on 301 osu! gameplay sessions. We report descriptive statistics, visualizations, correlations, and hypothesis-testing results.

A. Dataset Overview

Table III summarizes the numeric features of the dataset, including raw metrics (accuracy, max combo, hits), derived features (combo ratio, object density, miss rate), and temporal variables (day, hour).

TABLE III: Numeric Feature Summary

Feature	Count	Mean	Std	Min	25%	50%	75%	Max
accuracy	301	91.713	5.480	70.230	88.540	92.760	95.820	99.230
max_combo_player	301	353.452	227.062	68	189	293	452	1301
max_combo_beatmap	301	1196.505	572.397	402	745	1093	1520	3273
max_great	301	783.894	352.387	282	503	733	963	1880
great	301	655.625	306.573	86	411	597	821	1747
ok	301	75.668	68.369	2	23	54	112	427
meh	301	7.355	9.608	0	1	4	11	65
miss	301	17.256	13.367	0	7	14	24	68
ar	301	9.126	0.191	8	9	9	9.2	10
cs	301	3.990	0.233	3	4	4	4	5.7
bpm	301	179.167	22.736	87	170	175	190	320
hit_length	301	176.738	73.627	73	113	164	216	441
count_circles	301	462.781	238.625	149	294	422	545	1338
count_sliders	301	319.445	164.734	74	175	287	416	870
count_spinners	301	1.668	1.624	0	1	1	2	8
difficulty_rating	301	5.470	0.401	4.447	5.192	5.452	5.797	6.305
combo_ratio	301	0.333	0.213	0.041	0.172	0.269	0.421	1.000
miss_rate	301	0.022	0.016	0.000	0.010	0.020	0.031	0.100
total_objects	301	783.894	352.387	282	503	733	963	1880
object_density	301	4.443	0.782	2.946	3.971	4.309	4.748	7.667
hour	301	17.754	4.846	0	15	19	22	23
day	301	3.216	2.673	0	1	2	6	8

B. Univariate Analysis

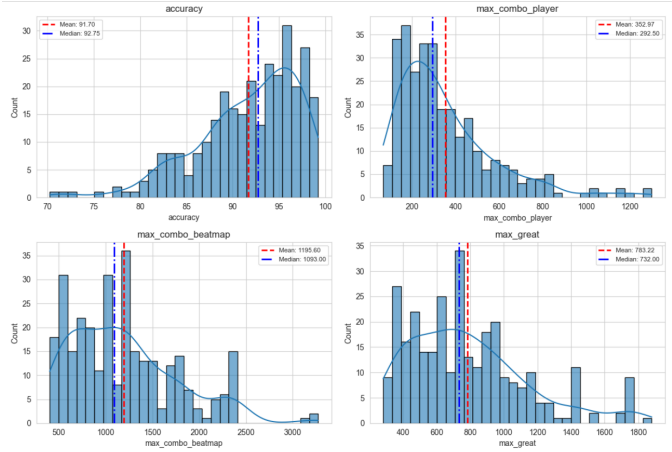

Fig. 1: Histogram distributions of accuracy, max combo player, max combo beatmap, and max great.

Histogram and KDE plots show that accuracy is slightly left-skewed, with most sessions clustering around 92–95%, a mean of 91.70%, and a median of 92.75%, indicating a few low-accuracy outliers. In contrast, max combo player exhibits a right-skewed distribution, with a mean of 352.97 and a median of 292.50, suggesting that most sessions have lower combos while a few high-performance sessions increase the average. Max combo beatmap similarly shows right-skewness, with a mean of 1195.60 and a median of 1093.00, reflecting variability in beatmap design and length. Max great also follows a right-skewed pattern, with a mean of 783.22 and a median of 732.00, indicating that while most sessions achieve moderate "great" counts, some high-performing sessions produce larger values (Fig. 1).

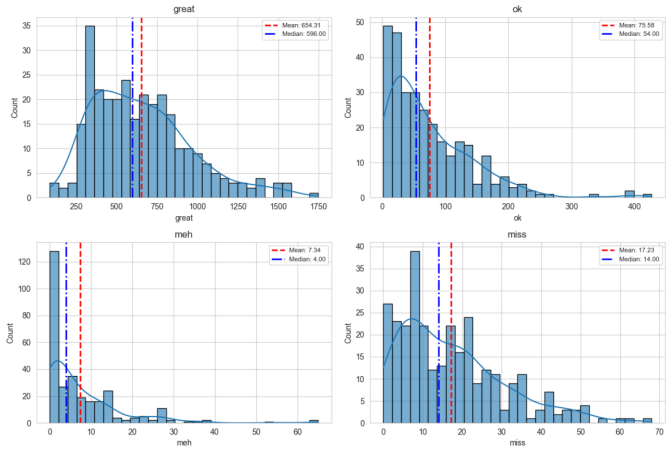

Fig. 2: Histogram distributions of great, ok, meh, and miss.

The distributions of hit accuracy metrics related to note judgments are predominantly right-skewed. The number of "great" hits shows a mean of 654.31 and a median of 656.00, indicating that most sessions cluster near the median with occasional very high counts. The "ok" hits display stronger right skewness, with a mean of 75.58 and a median of 54.00, suggesting that a few sessions contribute unusually high "ok" counts. Similarly, "meh" hits are right-skewed, with a mean of 7.34 and a median of 4.00, reflecting that most sessions have low "meh" counts while some outlier sessions register higher values. The miss count also follows a right-skewed pattern, with a mean of 17.23 and a median of 14.00, highlighting that most sessions have relatively low misses while a few poor-performance sessions increase the average (Fig. 2).

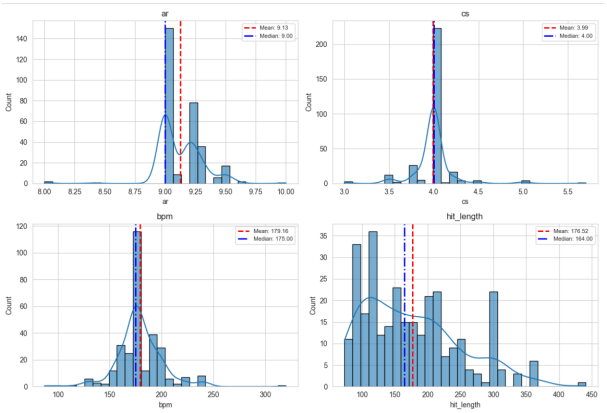

Fig. 3: Histogram distributions of Approach Rate, Circle Size, BPM, and Hit Length.

The distributions of beatmap-related features show varying patterns. Approach rate (AR) is roughly symmetric with a slight left skew, having a mean of 9.13 and median of 9.00, indicating most sessions cluster near the median with few higher-AR maps. Circle size (CS) is approximately symmetric but slightly right-skewed, with a mean of 3.99 and median of

4.00, showing most sessions involve medium-sized circles with occasional larger sizes. Beats per minute (BPM) exhibits a slight right skew, with a mean of 179.16 and median of 175.0, reflecting that most maps have moderate BPM values while some faster maps raise the mean. Hit length is right-skewed, with a mean of 176.52 and median of 164.00, suggesting that while most maps are of moderate length, a few long maps increase the average.(Fig. 3).

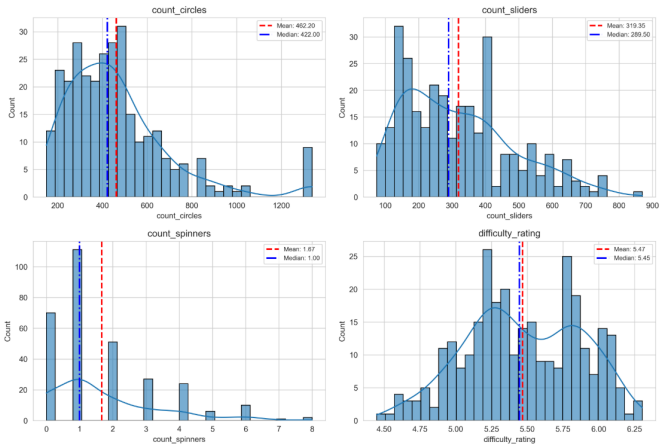

Fig. 4: Histogram distributions of Circle Count, Slider Count, Spinner Count, Difficulty Rating.

The distributions of map object counts and difficulty metrics are generally right-skewed. Circle count has a mean of 462.20 and median of 422.00, indicating most maps contain a moderate number of circles while a few maps with very high counts increase the mean. Slider count is also right-skewed, with a mean of 319.35 and median of 289.50, showing that most maps have a moderate number of sliders but some long maps elevate the average. Spinner count has a pronounced right skew, with a mean of 1.67 and median of 1.00, reflecting that most maps contain few or no spinners while occasional maps include several. Difficulty rating follows a similar pattern, right-skewed with a mean of 5.47 and median of 5.45, indicating that most maps are moderately difficult with some high-difficulty outliers.(Fig. 4).

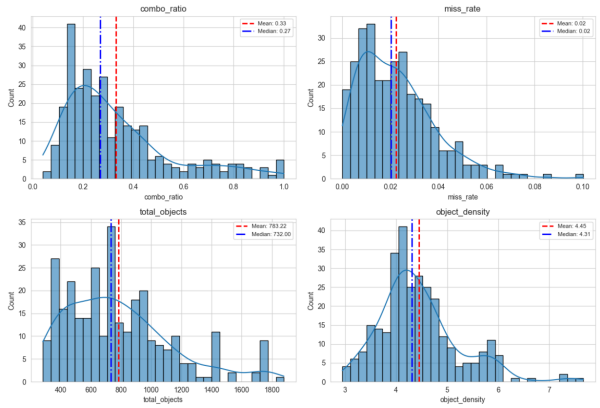

Fig. 5: Histogram distribution of Combo Ratio, Miss Ratio, Total Objects, and Object Density

Performance-related and map composition metrics exhibit right-skewed distributions. Combo ratio has a mean of 0.33 and median of 0.27, indicating that most sessions achieve moderate combo completion while a few sessions reach very high ratios. Miss rate is similarly right-skewed, with a mean and median both around 0.20, showing that most sessions have low miss occurrences but some have unusually high misses. Total objects follow a right-skewed pattern, with a mean of 783.22 and median of 732.00, reflecting that most maps contain a moderate number of objects while larger maps increase the average. Object density also shows a right-skew, with a mean of 4.45 and median of 4.31, suggesting most maps have moderate density but some maps are much more densely packed.(Fig. 5).

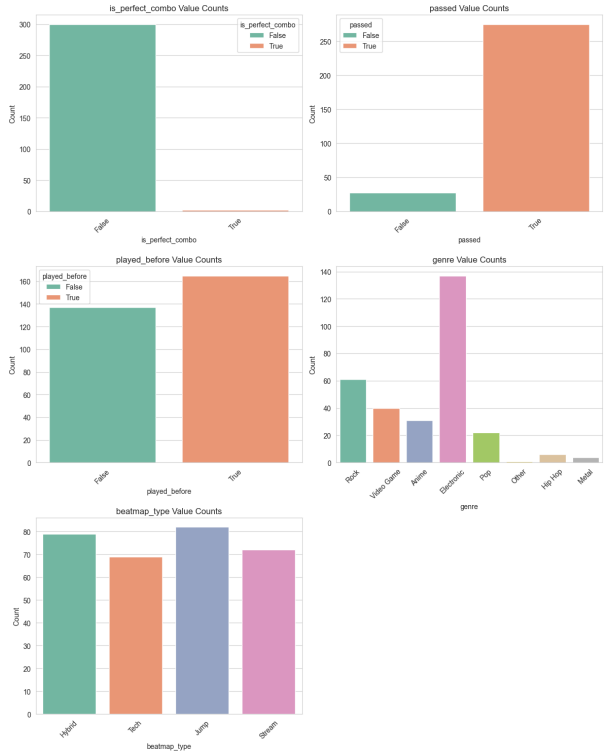

Fig. 6: Histogram distribution of Perfect Combo, Passed, Played, Genre, and Beatmap Type

The Bar Graph shows clear patterns in gameplay outcomes and map selection. For the boolean metrics, almost all sessions were not perfect combos, with only 2 out of 301 sessions achieving a perfect combo. Most sessions were passed (275), while 27 were failed, and the distribution of previously played maps is roughly balanced, with 165 sessions played before and 137 new maps. Among genres, Electronic is the most played (137 sessions), followed by Rock (61) and Video Game music (40), with smaller counts for Anime, Pop, Hip Hop, Metal, and Other. Beatmap types are fairly evenly distributed, with Jump (82) and Hybrid (79) slightly more common than Stream (72) and Tech (69), indicating the player experienced a variety of map types across sessions.(Fig. 6).

C. Temporal Analysis

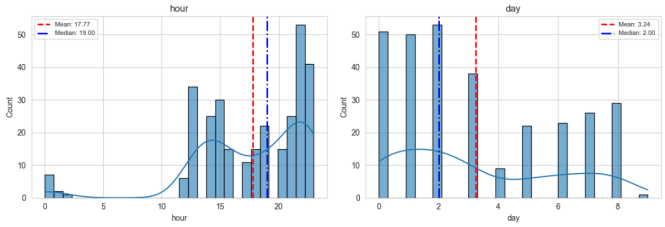

Fig. 7: Histogram distribution of Hours and Days

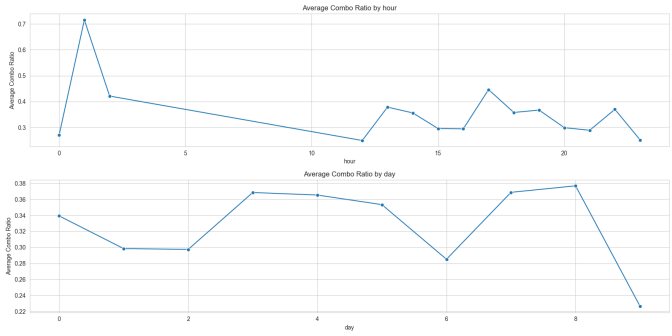

Fig. 8: Line Graph of Average Combo Ratio of Hours and Days

Temporal analysis of gameplay sessions reveals distinct patterns in the timing of play. The distribution of gameplay hours is slightly left-skewed, with a mean of 17.77 and a median of 19.00, indicating that most sessions occurred in the late afternoon to evening, while fewer sessions were recorded in the early morning. Daily session counts show a right-skewed distribution, with a mean of 3.24 and a median of 2.0, suggesting that most days had only a few gameplay sessions, with occasional days of higher activity. These trends highlight the player's preferred time-of-day and variability in daily gameplay frequency. (Fig. 7).

D. Bivariate Analysis

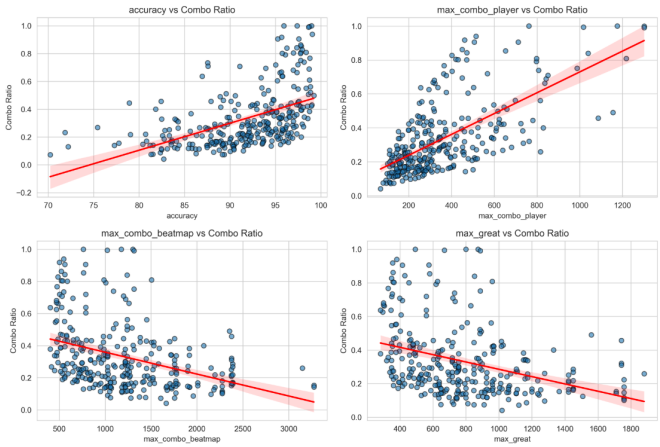

Fig. 9: Scatterplot of accuracy, max combo player, max combo beatmap, and max great.

Scatterplots show that accuracy positively correlates with combo ratio, with higher accuracy generally leading to higher combo ratios. Max combo player also displays a strong positive trend with combo ratio, indicating that sessions with higher maximum player combos correspond to higher performance. Conversely, max combo beatmap and max great show negative trends, suggesting that larger beatmap combos or more great hits in a beatmap do not always translate to higher player combo ratios, potentially reflecting increased map difficulty. (Fig. 9).

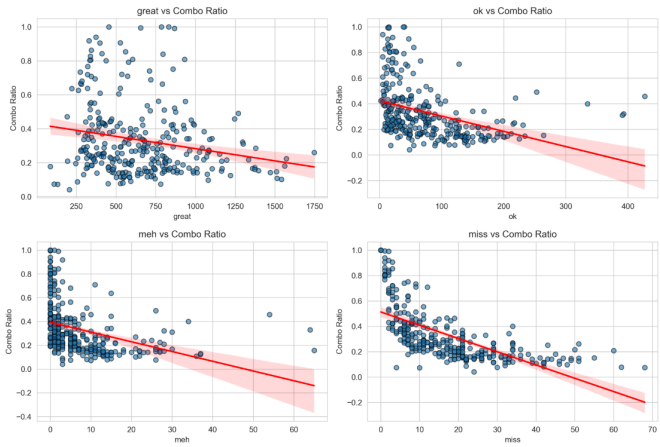

Fig. 10: Scatterplot of great, ok, meh, and miss.

The scatterplots indicate clear negative correlations between combo ratio and the counts of great, ok, meh, and miss hits. Higher numbers of ok, meh, and miss hits correspond to lower combo ratios, while great hits show a mild negative trend, highlighting the inverse relationship between performance errors and sustained combo execution.(Fig. 10).

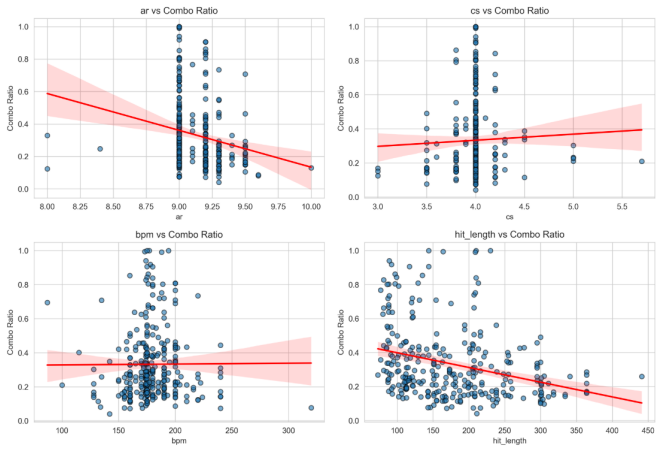

Fig. 11: Scatterplot of Approach Rate, Circle Size, BPM, and Hit Length.

For beatmap parameters, AR shows a slight negative correlation with combo ratio, suggesting that faster approach rates may reduce combo performance. CS and BPM appear largely uncorrelated, with points widely dispersed and the trendline nearly flat. Hit length exhibits a moderate negative trend, indicating that longer songs may be associated with lower combo ratios.(Fig. 11).

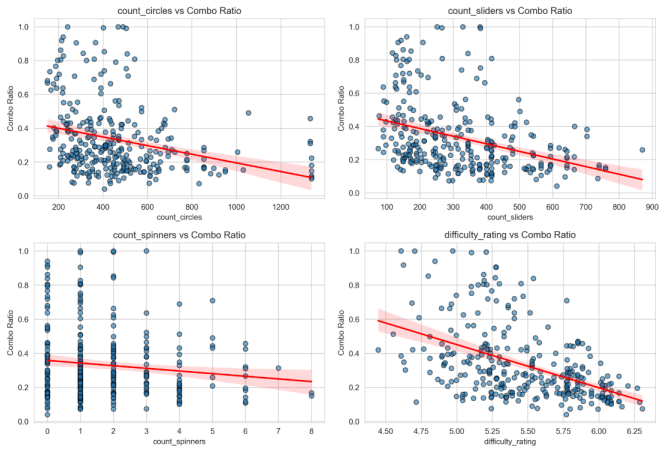

Fig. 12: Scatterplot of Circle Count, Slider Count, Spinner Count, Difficulty Rating.

Scatterplots reveal negative correlations between combo ratio and count_circles, count_sliders, count_spinners, and difficulty_rating. Higher object counts and higher difficulty ratings tend to correspond to lower combo ratios, reflecting the challenge of maintaining long combos on more complex beatmaps.(Fig. 12).

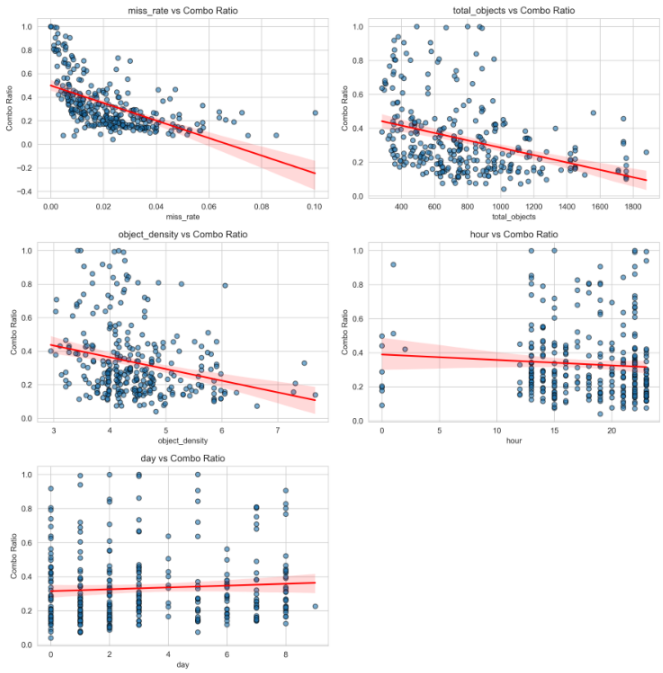

Fig. 13: Scatterplot of Miss Ratio, Total Objects, Object Density, Hours, and Days

The negative trends are pronounced for miss rate, total objects, and object density. Higher miss rates and greater numbers of total objects or object density are associated with lower combo ratios, consistent with the observation that more difficult or dense maps reduce sustained combo performance.(Fig. 13).

Scatterplots show a relatively flat trend between combo ratio and hour, suggesting that time of day does not significantly influence combo performance. Sessions occur mostly in the evening, but performance remains consistent across hours. Combo ratio shows a slight positive trend across days, with no extreme variations, indicating stable performance throughout the dataset timeframe.(Fig. 13).

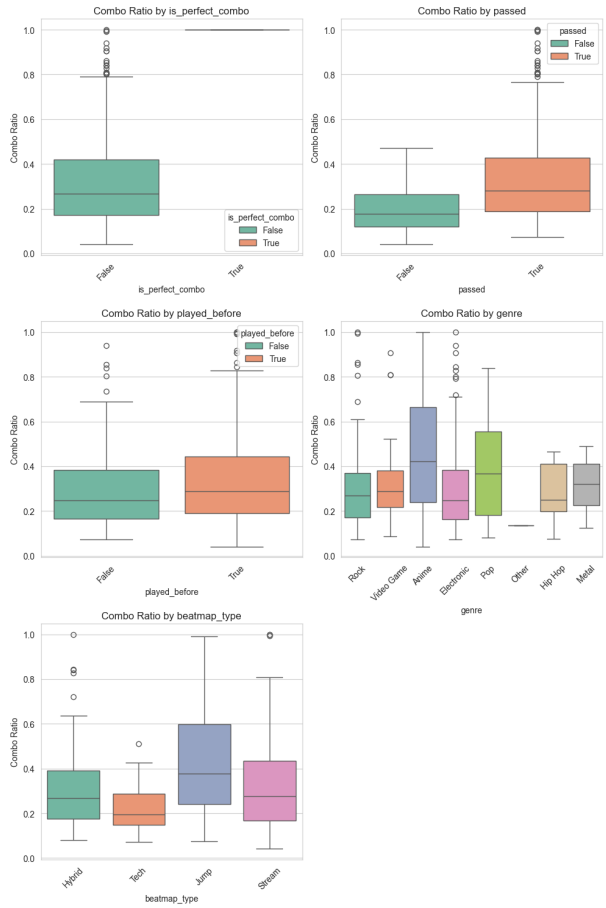

Fig. 14: Boxplot of Perfect Combo, Passed, Played, Genre, and Beatmap Type

Boxplot of categorical and boolean features highlights differences in combo performance across groups. For boolean variables, sessions with a perfect combo (True) show a mean combo ratio of 1.0 and zero variance, while non-perfect sessions (False) have a mean of 0.328 with a standard deviation of 0.206, indicating that perfect combos are rare but maximal when achieved. Passed sessions (True) show a higher mean combo ratio (0.345) compared to failed sessions (False, 0.200), reflecting better overall performance when passing. Similarly, previously played maps (True) exhibit a higher mean combo ratio (0.364) than new maps (False, 0.294), suggesting familiarity improves performance. Genre-wise, Jump maps have the highest mean combo ratio (0.432), followed by Video Game (0.328) and Rock (0.325), while Other shows the lowest (0.136), reflecting differences in player performance across music styles. For beatmap type, Jump maps again show the highest mean combo ratio (0.432), followed by Stream (0.347) and Hybrid (0.314), with Tech maps lowest (0.220), indicating that map design influences combo success. Overall, the boxplots reveal clear group-level differences in performance and variability, with certain map types, genres, and prior familiarity associated with higher combo ratios.(Fig. 14).

E. Correlation Matrix

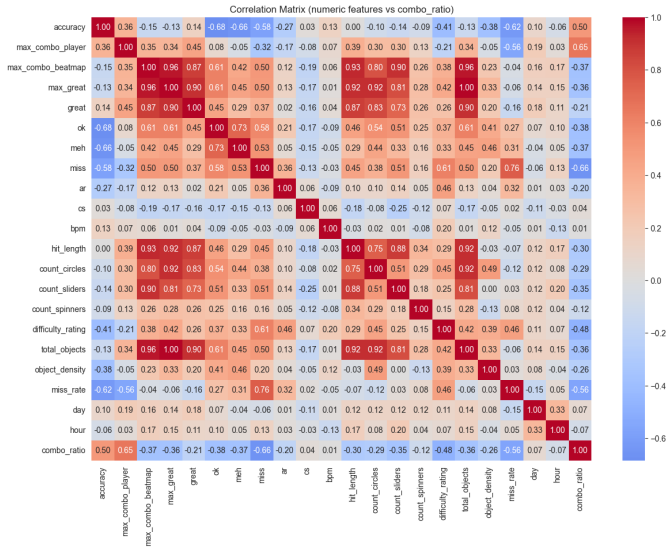

Fig. 15: Correlation Matrix of all numeric features

The correlation between `combo_ratio` and numeric features revealed several notable relationships. Strong positive correlations were observed for `max_combo_player` ($r = 0.654$) and `accuracy` ($r = 0.500$), indicating that precise execution and sustained combos enhance performance. In contrast, `miss` ($r = -0.658$), `miss_rate` ($r = -0.565$), and `difficulty_rating` ($r = -0.479$) showed strong-to-medium negative correlations, suggesting that errors and map complexity reduce `combo_ratio`. Other features, including `ok`, `meh`, `max_combo_beatmap`, `total_objects`, `object_density`, and `hit_length`, exhibited small-to-medium negative correlations, while `count_spinners`, `cs`, `bpm`, and temporal features (`day`, `hour`) showed negligible associations. Overall, the correlation matrix highlights that player performance and map complexity are the primary factors influencing `combo_ratio`, whereas temporal and minor map features have limited impact.

F. Statistical Tests

To examine the relationships between gameplay features and the player's performance, statistical tests were conducted across numeric, boolean, and categorical variables. Effect sizes are included to indicate the magnitude of observed effects.

1) Pearson Correlation: Pearson correlation coefficients were calculated to assess linear associations between numeric gameplay metrics and performance outcomes, such as combo ratio. Table **??** presents the correlation coefficients (r) along with p-values. The r-values also serve as effect sizes, interpreted as small (0.1–0.3), medium (0.3–0.5), or large (>0.5).

TABLE IV: Pearson correlation coefficients for numeric gameplay features (sorted by $|r|$)

Feature	r	p-value	Interpretation
miss	-0.658	<0.0001	Large Effect (-)
max_combo_player	0.654	<0.0001	Large Effect (+)
miss_rate	-0.565	<0.0001	Large Effect (-)
accuracy	0.500	<0.0001	Large Effect (+)
difficulty_rating	-0.479	<0.0001	Medium Effect (-)
ok	-0.383	<0.0001	Medium Effect (-)
meh	-0.371	<0.0001	Medium Effect (-)
max_combo_beatmap	-0.368	<0.0001	Medium Effect (-)
total_objects	-0.362	<0.0001	Medium Effect (-)
max_great	-0.362	<0.0001	Medium Effect (-)
count_sliders	-0.355	<0.0001	Medium Effect (-)
hit_length	-0.302	<0.0001	Medium Effect (-)
count_circles	-0.289	<0.0001	Small Effect (-)
object_density	-0.255	<0.0001	Small Effect (-)
great	-0.211	<0.0001	Small Effect (-)
ar	-0.205	<0.0001	Small Effect (-)
count_spinners	-0.119	0.0391	Small Effect (-)
day	0.072	0.2104	Negligible Effect (+)
hour	-0.071	0.2163	Negligible Effect (-)
cs	0.039	0.4973	Negligible Effect (+)
bpm	0.005	0.9286	Negligible Effect (+)

Interpretation: Misses, max_combo_player, and miss_rate show large correlations with combo ratio, highlighting their strong influence on sustained performance. Other metrics such as difficulty rating, ok, and meh demonstrate medium effects, while AR and object density show small effects.

2) Mann–Whitney U Test: Non-parametric Mann–Whitney U tests compared distributions of combo ratio across two groups. Effect sizes were calculated as:

$$r = \frac{Z}{\sqrt{N}}$$

TABLE V: Mann–Whitney U test results for boolean gameplay features

Feature	U-stat	p-value	Effect Size (r)	Interpretation
is_perfect_combo	598.000	0.0150	0.140	Small Effect
played_before	12,928.5	0.0243	0.130	Small Effect
passed	5,165.0	<0.0001	0.216	Small-to-Medium Effect

Interpretation: Achieving a perfect combo or playing previously familiar maps had small effects on combo ratio, while passing the map showed a slightly larger effect ($r = 0.216$), highlighting its stronger practical significance.

3) One-Way ANOVA (Categorical Features): One-way ANOVA tested differences in combo ratio across multiple groups for categorical variables. Effect size was reported as eta squared (η^2):

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}}$$

TABLE VI: One-way ANOVA results for categorical gameplay features

Feature	F-stat	p-value	η^2	Interpretation
beatmap_type	14.425	<0.0001	0.127	Large effect
genre	2.694	0.0102	0.060	Small-to-medium effect

Interpretation: Beatmap type strongly influenced combo ratio, with a large effect size, confirming that certain map types facilitate higher performance. Genre had a smaller but statistically significant effect, indicating minor differences in combo performance across music styles.

V. Discussion

This section interprets the findings from the analysis of 301 osu! gameplay sessions, explaining patterns in the data, relating them to prior research, addressing limitations, and proposing directions for future work.

A. Interpretation of Results

The analysis reveals several meaningful patterns in player performance.

1) Gameplay Metrics: For numeric gameplay performance features, Pearson correlation tests were conducted to examine linear relationships with `combo_ratio`. The null hypothesis (H_0) assumed no correlation ($r = 0$), while the alternative hypothesis (H_1) proposed a linear relationship ($r \neq 0$). Results strongly supported H_1 for several key metrics: `max_combo_player` ($r = 0.655$, $p < 0.0001$) and `accuracy` ($r = 0.500$, $p < 0.0001$) were positively correlated with `combo_ratio`, indicating that higher accuracy and sustained combo execution are associated with better performance. In contrast, performance-error metrics such as `miss` ($r = -0.656$, $p < 0.0001$), `miss_rate` ($r = -0.564$, $p < 0.0001$), `ok`, and `meh` were negatively correlated with `combo_ratio`, showing that mistakes significantly reduce the ability to maintain combos. For boolean features, Mann–Whitney U tests indicated that prior familiarity with a map (`played_before = True`, $r = 0.130$, $U = 12{,}928.5$, $p = 0.0243$), achieving a perfect combo (`is_perfect_combo = True`, $r = 0.140$, $U = 598.0$, $p = 0.0150$), and passing a map (`passed = True`, $r = 0.216$, $U = 5{,}165.0$, $p < 0.0001$) all significantly increased `combo_ratio`, highlighting the benefits of practice, successful execution, and map completion.

2) Gameplay Features: Beatmap characteristics were also evaluated for their influence on `combo_ratio`. Numeric features including `difficulty_rating` ($r = -0.477, p < 0.0001$), `object_density` ($r = -0.256$, $p < 0.0001$), and `hit_length` were negatively correlated with `combo_ratio`, reflecting the challenges of more complex or dense maps. Categorical features analyzed via one-way ANOVA revealed a strong effect of `beatmap_type` ($F = 14.425$, $\eta^2 = 0.127$, $p < 0.0001$), with Jump maps yielding the highest `combo_ratio`, followed by Stream and Hybrid maps, while Tech maps had the lowest. Music genre also had a smaller but significant effect ($F = 2.694$, $\eta^2 = 0.060$, $p = 0.0102$), with Electronic music supporting higher performance. These results confirm that both map design and musical style meaningfully influence gameplay outcomes.

3) Temporal Features: Temporal features, including `hour` and `day`, exhibited minimal correlation with `combo_ratio`, suggesting that the time at which a session occurs does not meaningfully affect player performance.

B. Comparison to Related Work

These results align with prior research showing that practice and prior exposure improve cognitive-motor coordination and timing accuracy [1]–[3]. The negative impact of map difficulty on performance supports earlier findings that increasing task complexity reduces sustained motor performance. Unlike some studies reporting time-of-day effects on cognitive tasks, our temporal analysis revealed minimal differences, likely due to the player's consistent evening gameplay routine. Additionally, the observation that certain map types and music genres facilitate higher performance provides new insight into how beatmap design interacts with individual skill that is an area less explored in prior work.

C. Limitations

This study has several limitations that may affect generalizability:

- Bias risk remains from map selection and session timing, despite objective logging.
- Short data collection window may miss longer-term trends.
- Missing contextual variables, such as fatigue, motivation, or distractions, could influence performance but were not captured.

D. Recommendations and Future Work

Future studies should address these limitations by:

- Conducting controlled experiments manipulating beatmap difficulty or session timing to isolate causal effects.
- Applying machine learning models to predict performance based on gameplay features, enabling automated identification of challenging maps or optimal practice strategies.
- Examining long-term longitudinal data to capture skill development and temporal trends.

VI. Conclusion

This study investigated how gameplay features and beatmap characteristics influence Combo Ratio in *osu!*, analyzing 301 personal gameplay sessions. The primary research question examined which factors affect sustained combo performance, while secondary questions considered differences across beatmap types, relationships with other performance metrics, and temporal patterns.

A. Key Findings

Players with higher `accuracy` and greater `max_combo_player` achieved higher `combo_ratio`, indicating that precise execution and sustained combo performance are critical for success. Conversely, more `misses`, higher `miss_rate`, and lower-quality judgments (e.g., `ok` and `meh`) were associated with lower `combo_ratio`, showing that mistakes significantly reduce sustained performance. Features such as `count_spinners` had weak but detectable effects, while `cs` and `bpm` had negligible influence. Additionally, prior experience

with a beatmap (played_before) positively impacted combo_ratio, emphasizing the benefits of practice and familiarity.

Denser, longer, and more complex maps modestly reduced performance. Features including difficulty_rating, total_objects, object_density, hit_length, max_combo_beatmap, count_sliders, count_circles, and ar were associated with lower combo_ratio, highlighting the impact of map complexity on player execution. Categorical factors also played a role: beatmap_type and music genre significantly influenced performance, with Jump maps and Electronic music corresponding to higher combo_ratio, while Tech maps and less familiar genres showed lower performance outcomes.

Time-related factors, including hour of play and day of the week, exhibited minimal correlations with combo_ratio, indicating that the timing of gameplay does not meaningfully affect sustained performance within the one-week dataset.

Overall, these findings suggest that player execution and map characteristics are the primary determinants of combo_ratio, while temporal factors have a minor influence.

B. Personal Insights

From this analysis, personal strengths were observed in previously played maps and genres such as Electronic and Jump, while weaknesses occurred in high-density, high-difficulty maps or unfamiliar genres. These insights confirm that both player-related factors (accuracy, prior experience) and beatmap characteristics (difficulty rating, object density, map type) significantly influence sustained Combo Ratio.

C. Real Applications

Practical implications include guiding targeted practice by focusing on unfamiliar or high-difficulty maps, leveraging preferred genres to maintain high Combo Ratio, and gradually increasing map complexity to improve overall performance. More broadly, this study demonstrates the value of personal data analysis in understanding individual behavior and skill development, with potential applications in other performance-based activities or learning environments.

In conclusion, the results answer the primary research question by identifying key performance determinants, while also providing actionable insights for personal improvement and strategies in rhythm game play. The combination of statistical evidence and self-observation emphasizes that both player skill and map characteristics jointly shape gameplay outcomes.

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